

GIS-OPTIMIZED LAST-MILE PARCEL LOCKER NETWORK DESIGN FOR A CITY: A MULTI-OBJECTIVE APPROACH

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CERTIFICATE

This is to certify that the report entitled "**GIS-Optimized Last-Mile Parcel Locker Network Design for a city: A Multi-Objective Approach**" is a bonafide record of the **Project I (ME3099E)** done by **Wadje Rushikesh Govindrao** (Roll No.: B231341ME), **Raihan S Basheer** (Roll No.: B231189ME), **Ron Sinoy** (Roll No.: B231209ME) and **Mohammed N V** (Roll No.: B231108ME) under my supervision, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Mechanical Engineering** from **National Institute of Technology Calicut**, and this work has not been submitted elsewhere for the award of a degree.

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ABSTRACT

Last-mile delivery is the most cost-intensive and environmentally burdensome segment of e-commerce logistics, contributing up to 53% of total supply chain cost. Parcel lockers offer a sustainable alternative by consolidating deliveries at secure collection points, eliminating failed delivery attempts, and reducing van kilometres. This project develops a **GIS-driven bi-level multi-objective optimization model** for the strategic placement and capacity planning of a parcel locker network in Indiranagar, Bangalore, a $3 \text{ km} \times 3 \text{ km}$ study area with 621 demand grid cells ($100 \text{ m} \times 100 \text{ m}$ each) and 20 candidate locker sites.

The bi-level framework models a Stackelberg game: the upper level minimizes the planner's total installation and operating cost while selecting locker locations and capacity types; the lower level maximizes aggregate customer utility, incorporating proximity, safety, transit access, and flexibility scores derived from a survey-based preference index. Three modelling fixes a slot turnover multiplier, a hard 500 m pedestrian walking constraint, and a K-sensitivity analysis over $K \in \{10, 15, 20\}$ sites are applied to produce a robust, deployable solution.

A **bipartite network analysis** is conducted to evaluate structural resilience: Dual Centrality (DC) and Coverage Centrality (CC) metrics identify the six irreplaceable locker sites, and a sequential disruption simulation shows that the network requires at least 16 of 17 open lockers to maintain 90% demand coverage. A 50-replication **SimPy discrete-event simulation** validates and subsequently rejects the MILP's assumed turnover of $r = 3$, finding a simulated effective turnover of $r_{\text{eff}} = 1.746 [1.714, 1.777]$ at 95% confidence, a 72% overestimate. The MILP is re-run with $r = 1.75$, revealing that the 20-site network achieves 48.9% demand coverage at 10% locker adoption with a first-attempt delivery success rate of 85.3%, saving an estimated **825 tonnes of CO₂ per year**.

Keywords: last-mile logistics, parcel lockers, bi-level optimization, discrete-event simulation, bipartite network analysis, GIS, Indiranagar, Bangalore.

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LIST OF ABBREVIATIONS

BILP	Binary Integer Linear Program
CBC	Coin-or Branch and Cut (open-source MILP solver)
CC	Coverage Centrality
CI	Confidence Interval
DC	Dual Centrality
DES	Discrete-Event Simulation
GIS	Geographic Information System
GPS	Global Positioning System
IT	Information Technology
KKT	Karush–Kuhn–Tucker (optimality conditions)
LP	Linear Program
MILP	Mixed-Integer Linear Program
NIT	National Institute of Technology
OSM	OpenStreetMap
QGIS	Quantum Geographic Information System
SEM	Standard Error of the Mean
SI	International System of Units
WGS	World Geodetic System

LIST OF SYMBOLS

Sets

G	Set of demand grid cells in the planning area
L	Set of candidate parcel-locker sites
T	Set of locker capacity types {Small, Medium, Large}
H	Set of hourly time slots (06:00–22:00)
F	Set of feasible (grid, locker) pairs within 500 m
N_g	Set of lockers reachable from grid g within 500 m

Parameters

$F_{t,l}$	Fixed installation cost of locker type t at site l (INR)
$O_{t,l}$	Operating cost per parcel for type t at site l (INR/parcel)
$C_{t,l}$	Physical slot capacity of locker type t at site l
$C_{eff_t,l}$	Effective daily throughput = $C_{t,l} \times r$ (parcels/day)
N_g	Locker-adoption demand from grid g (parcels/day)
D	Total daily demand across all grids (parcels/day)
$d_{g,l}$	Walking distance from grid g centroid to locker l (m)
δ	Maximum walking threshold: $\delta = 500$ m
$\tau_{g,l,h}$	Travel time from grid g to locker l at hour h (hours)
$p_{g,l}$	Preference index of locker l from grid g (dimensionless, 0–1)
r	Slot turnover multiplier (parcels/slot/day)
β	Proximity utility scaling constant ($\beta = 1$)
α_i	Utility weight for component i ($i = 1,2,3,4$; sum = 1)
S_l	Safety score of locker site l (0–1)
T_l	Transit access score of locker site l (0–1)

A_l Flexibility (24/7 access) score of locker site l (0–1)

Decision Variables

z_l Binary: 1 if locker site l is opened, 0 otherwise

$x_{t,l}$ Binary: 1 if type t installed at site l , 0 otherwise

$y_{g,l}$ Continuous: fraction of grid g demand assigned to locker l

Derived Metrics

$U_{g,l}$ Customer utility for assigning grid g to locker l

DC_l Dual Centrality of locker l (number of exclusively served grids)

CC_l Coverage Centrality of locker l (demand drop when l removed, p/d)

r_{eff} Effective turnover back-computed from simulation (parcels/slot/day)

FAR First-Attempt success Rate (%)

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The global e-commerce sector has experienced unprecedented growth over the past decade. In India alone, the e-commerce market was valued at USD 70 billion in 2023 and is projected to reach USD 350 billion by 2030 [2]. This growth has placed enormous pressure on urban last-mile delivery infrastructure, the final leg of the supply chain from a distribution hub to the customer's doorstep. Last-mile delivery accounts for up to **53% of total logistics cost** [3] and is responsible for a disproportionate share of urban traffic congestion, carbon emissions, and failed delivery attempts.

Bangalore, India's Silicon Valley, presents a particularly acute version of this challenge. Indiranagar, a 3 km × 3 km mixed residential-commercial neighbourhood housing a dense population of IT professionals, young urban residents, and a vibrant retail corridor, generates over 21,000 parcel deliveries per day. Studies report that 25–37% of home deliveries in Indian urban areas fail on the first attempt [4], requiring costly re-delivery trips that add both monetary cost and CO₂ emissions to the last-mile footprint.

Parcel lockers, secure, self-service electronic collection cabinets placed at convenient public locations such as Metro stations, supermarkets, and bus stops, offer a proven solution to the last-mile inefficiency problem. Ranieri et al. [5] demonstrated that adopting parcel lockers can reduce carbon emissions from last-mile delivery by up to 83% compared to home delivery. However, realizing these benefits requires **optimal placement**: a locker network that is too sparse leaves demand uncovered; one that is too dense wastes capital. The central research question of this project is therefore:

For a given set of candidate sites, what is the optimal number, location, and capacity type of parcel lockers in Indiranagar, Bangalore, such that planner costs are minimized and customer utility is maximized simultaneously?

1.2 PROBLEM DEFINITION

This project formulates and solves a **bi-level multi-objective optimization model** for parcel locker network design in Indiranagar. The model is a Mixed-Integer Linear Program (MILP) structured as a Stackelberg game: the logistics planner (leader) selects locker sites and capacity types to minimize total installation and operating cost; customers (followers) then choose preferred lockers from the opened set to maximize their aggregate utility.

The model accounts for the following considerations:

1. A hard 500 m pedestrian walking distance constraint, derived from Bangalore's urban heat and footpath conditions.
2. A customer preference index computed from survey-elicited top-2 locker preferences, following Dayal et al. [1].
3. Three locker capacity types (Small: 15 slots, Medium: 25 slots, Large: 35 slots) with heterogeneous fixed and operating costs.
4. A slot turnover multiplier r representing the number of parcel deliveries per physical slot per day, validated via discrete-event simulation.

Beyond the optimization model, the project includes: (i) a **bipartite network analysis** to quantify structural resilience; (ii) a **SimPy discrete-event simulation** with 50 replications to validate the turnover assumption and measure first-attempt delivery success rates; and (iii) **GIS visualisation** of all spatial results using Python geopandas.

1.3 SCOPE AND LIMITATIONS

The study covers the following scope:

- Study area: Indiranagar, Bangalore (approx. $3 \text{ km} \times 3 \text{ km}$).
- 621 demand grid cells of $100 \text{ m} \times 100 \text{ m}$ resolution.
- 20 candidate locker sites identified from field survey and OSM data.
- Synthetic demand dataset generated using Gaussian spreading calibrated to real Indiranagar land use and census data.
- Solver: Python PuLP with CBC backend (open-source).

Limitations of the study include the following:

- The dataset is synthetic; real field surveys would improve demand accuracy.
- The customer preference index is based on elicited preferences from a limited sample and does not account for sociodemographic heterogeneity.
- Travel times are computed using average speed profiles; actual GPS-traced travel times would improve the utility computation.
- The model does not account for dynamic demand variation across days of the week or seasonal fluctuations beyond the annual average.

1.4 OBJECTIVES

The specific objectives of this project are as follows:

1. Formulate a bi-level MILP for parcel locker location and capacity selection, considering a use case of Indiranagar at Bangalore, incorporating both planner cost minimization and customer utility maximization.
2. Apply three modelling fixes, turnover multiplier, hard walking constraint, and K-sensitivity analysis, improving upon the Phase 1 greedy heuristic.
3. Conduct a bipartite network analysis to identify structurally irreplaceable locker sites and quantify network resilience to sequential failures.
4. Validate the MILP turnover assumption via a 50-replication SimPy discrete-event simulation and revise the model accordingly.
5. Produce GIS maps of demand intensity, coverage buffers, assignment flows, resilience tiers, and land use overlays for the study area.
6. Quantify the environmental impact in terms of CO₂ savings of the optimal locker network compared to the baseline home-delivery scenario.

1.5 OUTLINE OF THE REPORT

The report is organized as follows.

Chapter 2 reviews the existing literature on last-mile logistics, parcel locker optimization, bi-level programming, and simulation-based validation. **Chapter 3** describes the study area, data sources, and the synthetic dataset used in this project. **Chapter 4** presents the complete mathematical formulation of the bi-level MILP, including the upper-level planner model, the

lower-level customer model, the bipartite network analysis metrics, and the simulation methodology. **Chapter 5** reports and discusses all results: optimization outcomes, K-sensitivity, bipartite resilience analysis, simulation validation, corrected MILP re-run, and GIS maps. **Chapter 6** presents the conclusions, policy implications, and scope for future work.

CHAPTER 2

REVIEW OF LITERATURE

2.1 LITERATURE REVIEWED IN THE STUDY

Dayal, et al., (2025) proposes a bi-level optimization model for determining the optimal locations and capacities of parcel lockers in urban last-mile delivery systems. The model simultaneously considers the objectives of logistics planners and customer preferences, where the upper level minimizes setup and operational costs while the lower level maximizes customer utility based on travel time and preferred locker locations. The study uses survey data collected from 343 respondents in Mumbai and evaluates feasible locker locations such as railway stations, bus depots, malls, offices, and local markets. The results showed that only 11 parcel lockers were sufficient out of 37 feasible locations to satisfy customer demand efficiently. Around 81% of customers were allocated to one of their top two preferred locker choices, proving that incorporating customer preferences improves both operational efficiency and user satisfaction. The study highlights parcel lockers as an effective solution for reducing failed deliveries, lowering last-mile delivery costs, and supporting sustainable urban logistics.

Corradini, et al., (2025) presents a network science-based approach for optimizing parcel locker placement in urban last-mile delivery systems. Unlike traditional facility location models that mainly focus on minimizing cost or distance, the study models parcel locker systems as multilayer bipartite networks connecting demand zones and locker locations. The authors introduce two novel Social Network Analysis metrics, namely Dual centrality and Coverage centrality, to identify lockers that are not only highly accessible but also structurally important for maintaining network connectivity and resilience. The framework was applied to the cities of Milan, Rome, and Naples using open-source demographic and geographic data. The results showed that conventional coverage-based placement strategies often prioritize redundant high-demand hubs, whereas the proposed network-centric metrics identify hidden but critical lockers that act as bridges between different user communities. The study found that considering network connectivity and resilience alongside accessibility can improve the robustness of urban parcel locker systems. The paper highlights the importance of integrating complex network analysis into urban

logistics planning to develop more resilient, efficient, and equitable last-mile delivery infrastructure.

Sawik et al., (2022) proposes a multi-criteria simulation-optimization framework for planning Automated Parcel Locker (APL) networks in urban last-mile delivery systems. Using a real-world case study from Poznań, Poland, the authors combine agent-based simulation and mathematical optimization to determine the optimal number and placement of parcel lockers while minimizing setup, maintenance, and operational costs. The study also analyzes the growth of population, e-shoppers, APL users, and parcel demand over a three-year period. The results showed a significant increase in parcel demand and corresponding locker requirements, with the number of lockers growing from 52 to 122 over three years. The study highlights that efficient parcel locker allocation can reduce failed deliveries, improve last-mile delivery efficiency, and support sustainable urban logistics while controlling operational costs.

Hovi and Bø, (2024) analyzes the efficiency and environmental impact of parcel lockers compared to home deliveries and traditional pick-up points in e-commerce last-mile logistics. The authors developed a Decision Support Tool (DST) using real operational data from Norway to evaluate fixed costs, operational costs, delivery efficiency, and CO₂ emissions across different delivery methods. The study found that parcel lockers significantly improve last-mile efficiency by enabling consolidated deliveries, reducing failed delivery attempts, and lowering operational costs compared to attended and unattended home deliveries. The results showed that home deliveries were the most expensive and environmentally intensive delivery option, while parcel lockers became increasingly cost-effective and sustainable as the number of parcels delivered per locker increased. The study also highlighted that the environmental benefits of parcel lockers depend heavily on customer travel behavior during parcel collection, emphasizing the importance of strategically locating lockers within walking distance or along daily travel routes.

Hagberg et al., (2008) introduces NetworkX, a Python-based open-source framework developed for analyzing and modeling complex networks. The study explains how NetworkX supports graph creation, network analysis, shortest path calculations, clustering, centrality analysis, and visualization of large-scale network structures. The framework is

designed to handle complex network data efficiently and integrates with scientific computing libraries such as NumPy, SciPy, and Matplotlib for advanced analysis and visualization. The paper also demonstrates the application of NetworkX in studying synchronization and structural behavior in dynamic networks. The framework enables researchers to model and analyze real-world network systems efficiently, making it useful for applications involving transportation systems, logistics networks, urban infrastructure planning, and complex supply chain connectivity analysis.

Sabzevari Zadeh et al., (2026) proposes a strategic optimization framework for designing parcel locker networks in urban last-mile delivery systems. The study introduces the Parcel Locker Location-Routing Problem (PL-LRP), which simultaneously considers parcel locker location selection, capacity planning, fleet sizing, customer assignment, replenishment scheduling, inventory management, and vehicle routing within a single optimization model. Using a mixed-integer linear programming formulation and a Variable Neighborhood Search-based solution approach, the study demonstrates through a Chicago case study using Amazon delivery data that incorporating operational costs and flexible replenishment planning into network design significantly improves delivery efficiency, reduces long-term costs, and enhances the sustainability of parcel locker-based last-mile logistics.

Lachapelle et al., (2018) paper examines the location, accessibility, and urban planning implications of parcel locker systems in South East Queensland, Australia. Through micro-level site analysis and macro-level regional analysis, the study evaluates how parcel lockers are integrated into urban environments and their impact on accessibility, transportation, and sustainability. The findings show that most parcel lockers are located in automobile-oriented areas such as shopping centres, commercial streets, and Australia Post locations, with strong car accessibility but limited support for public and active transport modes. The study concludes that while parcel lockers improve delivery convenience and reduce failed deliveries, their current deployment pattern in car-dominant cities does not fully support sustainable urban transport goals. The paper highlights the importance of strategically locating parcel lockers near transit hubs, mixed-use developments, and high-density areas to improve accessibility, reduce travel distances, and better integrate parcel locker systems into sustainable urban logistics planning.

Moslem and Pilla (2023) proposes a hybrid decision-support framework using the Spherical Fuzzy Analytic Hierarchy Process (SFAHP) to identify the most suitable parcel locker locations in Dublin, Ireland. The research evaluates six possible location alternatives based on five major criteria: reliability, accessibility, traffic and operations, security, and environmental impact. Findings reveal that post offices emerged as the most suitable parcel locker locations, followed by public transport stops and private car parking areas, while pedestrian areas were ranked least suitable. The study highlights that integrating fuzzy multi-criteria decision-making methods can effectively support policymakers and urban logistics planners in optimizing parcel locker placement for improved last-mile delivery efficiency, accessibility, and sustainability.

Vakulenko et al., (2018) explores customer perceptions and value creation associated with parcel lockers in e-commerce last-mile delivery systems using focus group studies conducted in Sweden. The study identifies four major types of customer value generated by parcel lockers: functional value (convenience, speed, accessibility), emotional value (comfort and control), social value (reduced or preferred human interaction), and financial value (cost savings). The findings show that parcel lockers improve delivery flexibility and customer convenience, but customer satisfaction strongly depends on service reliability, ease of use, accessibility, and previous user experience. The study also highlights that parcel lockers shift part of the delivery responsibility to customers, making users active participants in the value creation process within self-service logistics systems.

Zhang and Demir, (2025) presents a systematic literature review of parcel locker (PL) solutions in last-mile delivery, focusing on operational efficiency, sustainability, and future research directions. The study reviews 86 research articles published between 2017 and 2025 and highlights that most existing research is concentrated on parcel locker location optimization, routing strategies, network design, and cost reduction. The review also examines the environmental and social impacts of parcel lockers, including reduced emissions, lower traffic congestion, improved delivery efficiency, and enhanced customer convenience. Additionally, the paper identifies major research gaps such as limited studies on mobile parcel lockers, integration with smart urban logistics systems, and the use of AI-driven technologies in parcel locker operations, while proposing future directions for developing more sustainable and intelligent last-mile delivery systems.

2.2 Research Gaps Identified

Based on the reviewed literature, several important research gaps were identified in the existing parcel locker and last-mile delivery studies. Most studies focus primarily on minimizing operational cost and travel distance, while limited attention is given to integrating real-world customer behavior, dynamic demand variations, environmental sustainability, and network resilience within a single framework.

A major gap observed in the literature is the heavy dependence on static models. Studies such as Dayal et al. (2025), Corradini et al. (2025), and Moslem & Pilla (2023) use fixed demand patterns, static traffic conditions, and predetermined customer behavior, which do not reflect the highly dynamic nature of Indian urban logistics systems. Real-time demand fluctuations, hourly traffic variability, seasonal peaks, and live locker occupancy are largely ignored in existing models.

Another important gap is the lack of integration between optimization, sustainability, and resilience analysis. While several studies optimize locker placement or routing efficiency, only a limited number incorporate carbon emissions, failed deliveries, and environmental externalities into the optimization objectives. Similarly, network resilience against disruptions such as traffic congestion or weather-related failures is rarely analyzed despite its growing importance in dense urban delivery systems.

The literature also shows limited focus on the Indian urban context. Most existing studies are based on European, Chinese, or North American cities, where travel behavior, infrastructure quality, and customer accessibility patterns differ significantly from Indian cities. There is very limited work combining GIS-based urban analysis, customer utility, carbon emissions, and parcel locker optimization specifically for dense Indian urban regions such as Bengaluru.

In addition, many existing studies treat parcel lockers as fixed and isolated infrastructure systems without considering modular locker configurations, dynamic capacity utilization, shared open-access networks, or interoperability between logistics providers. Research integrating bipartite network analysis, bi-level optimization, and discrete-event simulation into a unified decision-making framework is also highly limited in the current literature.

2.3 Summary of Literature Review and Research Gaps

The reviewed literature shows that parcel locker systems are increasingly being recognized as an effective solution for improving last-mile delivery efficiency, reducing failed deliveries, lowering operational costs, and supporting sustainable urban logistics. Existing studies mainly focus on optimizing parcel locker locations, routing strategies, customer allocation, operational cost reduction, and network design using approaches such as bi-level optimization, mixed-integer programming, simulation-optimization, social network analysis, and multi-criteria decision-making models. Several studies also highlight the environmental and customer-related benefits of parcel lockers, including reduced CO₂ emissions, improved delivery flexibility, enhanced customer convenience, and better delivery success rates.

Table 2.1. Comparison of the present study with Dayal et al. (2025)

Aspect	Dayal et al. (2025) — Mumbai	Present Study — Indiranagar, Bangalore
Study area	Mumbai suburban corridor	Indiranagar, Bangalore (3 km × 3 km)
Grid cells	100 grids (500 m ²)	621 cells (100 m × 100 m)
Candidate sites	37 locations	20 real sites
Traffic speed	Static 23 km/h	Hourly τ : 8–23 km/h
Walking constraint	Not modelled	Hard δ = 500 m cap
Customer utility	$U = \beta/\tau$ (proximity only)	$U = \alpha_1(\beta/\tau) + \alpha_2S_i + \alpha_3T_i + \alpha_4A_i$
Upper-level cost	Fixed + Operating (2 terms)	Fixed + Operating (4-term w/ K constraint)
Solver	BilevelJuMP (Julia)	Python PuLP (CBC)

However, the literature reveals several significant research gaps. Most existing models rely on static demand patterns, fixed traffic conditions, and simplified customer behavior assumptions, without capturing the dynamic nature of real-world urban logistics systems. Environmental sustainability and network resilience are often studied separately rather than being integrated directly into optimization frameworks. Additionally, limited research exists on real-time locker occupancy, dynamic routing, modular locker configurations, and shared open-access locker networks. Most studies are also concentrated in Western, European, or Chinese urban contexts, leaving a major gap in research focused on dense Indian urban environments such as Bengaluru. Furthermore, very few studies

combine GIS-based spatial analysis, customer utility modeling, bipartite network analysis, and discrete-event simulation within a single integrated framework for parcel locker optimization. A comparison between the present study and Dayal et al. is tabulated in Table 2.1.

CHAPTER 3

AREA, DATA AND SYNTHETIC DATA

3.1 STUDY AREA: INDIRANAGAR, BANGALORE

Indiranagar is a mixed residential-commercial neighbourhood in eastern Bangalore (12.970–12.993°N, 77.630–77.654°E), covering approximately 9 km² of which the project focuses on a 3 km × 3 km core (9 km²). The neighbourhood is home to a dense population of IT professionals and urban residents associated with the surrounding technology parks (Domlur IT Park, HAL Aerospace Complex, and the Outer Ring Road IT corridor). The presence of two Namma Metro stations (Indiranagar and Domlur on the Purple Line), a dense retail corridor along 100 Feet Road and CMH Road, and a high density of food-delivery and e-commerce consumers makes Indiranagar one of the highest-volume last-mile delivery zones in Bangalore.

The study area is bounded by Ulsoor Lake to the north (a natural demand gap), the HAL Airport Road to the east, Old Madras Road to the south, and the Domlur flyover to the west. This boundary captures the full demand catchment of the Indiranagar retail and residential core while excluding the lower-density industrial zones on the periphery.

3.2 Demand Grid Construction

The 3 km × 3 km study area is partitioned into a regular grid of 100 m × 100 m cells (27 columns × 23 rows = 621 cells), each identified by the GPS coordinates of its centroid. The 100 m resolution was chosen following Marcucci et al. (2021) as the minimum spatial unit consistent with individual building-block demand variation in a high-density urban neighbourhood. Coarser resolution (e.g., 500 m) would average out the demand peaks associated with individual IT parks and apartment complexes; finer resolution (e.g., 50 m) would introduce noise from survey sparsity.

Each grid cell g is associated with a daily parcel demand N_g (parcels/day), computed as described in Section 3.4, and a land use classification derived from the Bruhat Bengaluru Mahanagara Palike (BBMP) zoning map: residential, commercial, mixed-use, institutional, industrial, or parks/open space.

3.3 Candidate Locker Sites

Twenty candidate locker sites were identified through a combination of field survey, OSM data, and domain knowledge of the Indiranagar area. Site selection criteria followed Iwan et al. (2016): sites were required to be (i) publicly accessible on foot; (ii) located adjacent to a high-footfall anchor (transit stop, supermarket, pharmacy, or apartment complex gate); (iii) capable of accommodating a standard locker cabinet (1.8 m × 0.6 m footprint) in a covered or sheltered location; and (iv) connected to the electrical grid. The 20 sites span all major sub-zones of the study area: the 100 Feet Road commercial corridor (L01–L04), the Domlur-CMH cluster (L05–L09), the HAL-Kodihalli residential zone (L09–L12, L16–L17), the Stage 1–2 residential belt (L08, L13–L15), and the peripheral fringe sites (L18–L20). Table 3.1 presents the 20 candidate locker sites with spatial scoring attributes.

Table 3.1. Candidate locker sites with spatial scoring attributes.

ID	Site Name	Type	Safety Score	Transit Score	Flex Score	24/7
L01	Indiranagar Metro Station	Transit	1.00	1.00	0.90	Yes
L02	CMH Road Bus Stop	Transit	0.85	0.90	0.70	No
L03	100 Feet Rd Supermarket	Retail	0.90	0.80	0.80	No
L04	12th Main Rd Coffee Shop	Retail	0.80	0.70	0.65	No
L05	Defense Colony Park Entrance	Public	0.85	0.65	0.90	Yes
L06	Domlur IT Park Gate	Institutional	0.90	0.75	0.60	No
L07	Indiranagar Post Office	Government	0.85	0.75	0.70	No

L08	Stage 1 Apartment Complex	Residential	0.90	0.65	1.00	Yes
L09	HAL 2nd Stage Shopping Centre	Retail	0.80	0.70	0.70	No
L10	Kodihalli Metro Feeder Stop	Transit	0.90	0.95	0.80	No
L11	Binnamangala Convenience Store	Retail	0.75	0.60	0.75	No
L12	Old Madras Rd Petrol Bunk	Fuel/Retail	0.80	0.70	1.00	Yes
L13	Indiranagar Stage 2 Market	Retail	0.85	0.80	0.75	No
L14	CMH Rd Pharmacy	Retail	0.80	0.80	0.80	No
L15	Domlur Layout Apartment Gate	Residential	0.85	0.65	1.00	Yes
L16	HAL Colony Club	Institutional	0.80	0.60	0.70	No
L17	Kodihalli Village Main Rd	Retail	0.70	0.60	0.70	No
L18	Richmond Town Fringe Superstore	Retail	0.85	0.75	0.80	No
L19	Domlur Bridge End	Transit/Public	0.80	0.70	0.85	Yes
L20	Ulsoor Lake Road Stop	Public	0.75	0.65	0.90	Yes

3.4 Synthetic Demand Dataset

Because real parcel delivery data for Indiranagar is commercially sensitive and unavailable, a synthetic demand dataset was generated using a Gaussian spreading kernel calibrated to publicly available land use and census data. This approach follows Marcucci et al. (2021) and is standard practice in urban logistics simulation studies where real data are unavailable.

Twelve demand hotspots were identified from Bangalore city census (2011, ward-level population density), the BBMP commercial zoning map, and field observation: the Indiranagar Metro station interchange, the 100 Feet Road–CMH Road commercial corridor, the Domlur IT Park employee zone, the HAL Aerospace staff residential colony, and eight residential apartment clusters identified from Google Maps review counts as proxies for delivery volume. The daily parcel demand for grid cell g is:

$$N_g = \sum_k w_k \cdot \exp\left(-\frac{d_{g,k}^2}{2\sigma_k^2}\right) \quad (3.1)$$

where $d_{g,k}$ is the Euclidean distance from the centroid of cell g to hotspot k , σ_k is the spread parameter (50–150 m depending on hotspot type), and w_k is the hotspot weight (calibrated so that total demand $D = \sum_g N_g = 21,984$ parcels/day, consistent with Indiranagar's estimated daily delivery volume at a 10% e-commerce penetration rate applied to the 2011 population of 218,000 residents and 15,000 daily IT park workers).

The resulting demand surface ranges from 5 parcels/day per cell in the Ulsoor Lake peripheral zone to 75 parcels/day per cell along the 100 Feet Road–CMH Road corridor. The spatial pattern is consistent with the BBMP commercial zoning map and with the qualitative field observations conducted during the site survey.

3.5 Locker Cost Parameters

Three locker capacity types are modelled, consistent with the product specifications of Parcel Pending (India) Pvt. Ltd. and Amazon Locker India:

Table 3.2. Locker capacity types and cost parameters.

Type	Physical Slots	Fixed Installation Cost (INR)	Operating Cost (INR/parcel)	Footprint
Small	15	1,87,000	3.50	0.9 m × 0.6 m
Medium	25	3,12,000	3.20	1.4 m × 0.6 m
Large	35	3,70,600 (avg)	2.90	1.8 m × 0.6 m

Fixed installation costs include civil works (foundation, enclosure), electrical connection, network connectivity (4G SIM), and the locker hardware itself. Operating costs are per parcel delivered and include maintenance, cleaning, cellular data charges, and amortised software licence fees. The Large type is cost-dominant on a cost-per-slot basis (INR 5,700–6,200 per slot versus INR 7,125 per slot for both Small and Medium), making it the optimal choice under any cost-minimizing objective. The locker capacity types and their associated cost parameters are summarised in Table 3.2.

3.6 Distance and Travel Time Matrices

The walking distance matrix $d_{g,l}$ (621×20) was computed from the GPS centroids of demand grid cells and locker sites using the Haversine formula, as an approximation to actual footpath distance (footpath circuit factor of 1.1–1.3 in Indiranagar is neglected for consistency with the 500 m hard constraint). The feasible set $F = \{(g, l) : d_{g,l} \leq 500 \text{ m}\}$ contains 881 grid-locker pairs, representing 35% of the 12,420 theoretically possible pairs.

Travel time $\tau_{g,l,h}$ for the preference utility function was computed from $d_{g,l}$ using hourly average speeds on the Indiranagar road network derived from Google Maps speed data (March 2024): peak-hour speed (16:00–19:00) of 8 km/h on the 100 Feet Road corridor, dropping to 6 km/h on CMH Road, with off-peak speed of 23 km/h. The peak demand hour $h^* = 17:00$ (the hour with the highest normalized demand fraction, $f_{17} = 0.1296$) is used as the reference hour in the lower-level utility computation.

3.7 Customer Preference Index

The preference index $p_{g,l}$ represents the relative attractiveness of candidate locker site l for customers residing or working in grid cell g . In the absence of primary survey data, which

lies beyond the scope of this B.Tech project, the preference index is **synthetically constructed** from objective accessibility and land-use attributes available in the dataset, following the structural role defined by Dayal et al. [1].

The two locker sites scoring highest on this composite for each grid are designated as its top-two preferred sites. The preference index is then computed as:

$$P_{g,l} = \text{number of top-2 preference slots assigned to locker } l \text{ from grid } g / 2$$

This yields $p_{g,l} \in \{0, 0.5, 1.0\}$ in $\{0, 0.5, 1.0\}$ for the ranked lockers, and a small fallback value of $p_{g,l} = 0.05$ for all other feasible but non-preferred pairs within 500 m. The fallback prevents the lower-level LP from being infeasible while still strongly penalising assignment to unattractive sites.

CHAPTER 4

MATHEMATICAL FORMULATION AND METHODOLOGY

This project formulates the parcel locker network design problem as a bi-level Mixed-Integer Linear Program (MILP), structured as a Stackelberg game between a logistics planner (leader, upper level) and customers (followers, lower level). The planner first decides which of the 20 candidate sites to open and what locker capacity type to install at each site; customers then choose their preferred locker from the open set to maximize aggregate utility. The bi-level structure ensures that the planner's cost-minimizing decisions are consistent with rational customer behaviour, a requirement absent in single-level formulations.

The model is solved by reformulating the bi-level program into a single-level MILP using the Karush–Kuhn–Tucker (KKT) optimality conditions of the lower-level LP, following the methodology of Dayal et al. (2025). The resulting MILP is solved using the Python PuLP library with the CBC open-source branch-and-cut solver.

4.1 Bi-Level Optimization Framework

The model is a **bi-level Mixed-Integer Linear Program (MILP)**. In the Stackelberg game sense, the logistics planner is the leader and customers are the followers. The planner first decides which sites to open and what capacity type to install; customers then choose their preferred locker from the open set.

1. **Upper level (Planner):** minimizes total cost, fixed installation + variable operating + van routing + overflow penalty.
2. **Lower level (Customers):** maximizes aggregate composite utility, proximity, safety, and flexibility subject to the planner's binary decisions.

4.1.1 Upper-Level Model

Objective Function

The planner minimizes total operational cost W :

$$\min W = \underbrace{\sum_{l \in \mathcal{L}} F_l z_l}_{\text{(i) Fixed cost}} + \underbrace{\sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{L}} x_{t,l} O_{t,l}}_{\text{(ii) Operating cost}} + \underbrace{\sum_{g \in \mathcal{G}} \sum_{l \in \mathcal{L}} R_{g,l} N_g y_{g,l}}_{\text{(iii) Routing cost}} + \underbrace{\sum_{l \in \mathcal{L}} \pi_l s_l}_{\text{(iv) Overflow penalty}} \quad (4.1)$$

Explanation of each term:

Fixed installation cost: $\sum_l F_l z_l$ sums the one-time capital cost over all opened locker sites. Different locations have different FI values reflecting land lease and civil work costs.

Variable operating cost: $\sum_t \sum_l x_{t,l} O_{t,l}$ accumulates the periodic running cost per locker, differentiated by both capacity type and site. Electricity, maintenance, cleaning, and remote-monitoring charges are included.

Van routing cost: $\sum_g \sum_l R_{g,l} N_g y_{g,l}$ represents the last-mile delivery cost. For every parcel ($N_g y_{g,l}$) routed from grid g to locker l , the per-parcel routing cost $R_{g,l}$ is incurred. This term incentivises the planner to open lockers close to dense demand grids.

Overflow penalty: $\sum_l \pi_l s_l$ penalizes unmet demand at each locker. Since π_l is large (M), the solver drives $s_l \rightarrow 0$, effectively enforcing capacity constraints softly and allowing infeasibility detection.

Constraints

$$\sum_{t \in \mathcal{T}} x_{t,l} \leq z_l, \quad \forall l \in \mathcal{L} \quad (4.2)$$

$$\sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{L}} x_{t,l} C_{t,l} \geq D \quad (4.3)$$

$$d_{g,l} \cdot y_{g,l} \leq \delta \cdot z_l, \quad \forall g \in \mathcal{G}, \forall l \in \mathcal{L} \quad (4.4)$$

$$s_l \geq \sum_{g \in \mathcal{G}} N_g y_{g,l} - \sum_{t \in \mathcal{T}} C_{t,l} x_{t,l}, \quad \forall l \in \mathcal{L} \quad (4.5)$$

$$s_l \geq 0, \quad \forall l \in \mathcal{L} \quad (4.6)$$

$$z_l \in \{0, 1\}, \quad \forall l \in \mathcal{L} \quad (4.7)$$

$$x_{t,l} \in \{0, 1\}, \quad \forall t \in \mathcal{T}, \forall l \in \mathcal{L} \quad (4.8)$$

Explanation of each constraint

(4.2) **Locker-type activation.** A capacity type can only be chosen at location l if that location is opened ($z_l = 1$). The summation also enforces that at most one capacity type is selected per site.

(4.3) **Aggregate demand coverage.** The total installed capacity across all selected lockers must satisfy the full planning area demand D . This is the system-level capacity adequacy constraint.

(4.4) **500 m walking distance constraint (novel contribution).** Grid g cannot be assigned ($y_{g,l} > 0$) to locker l unless (a) the locker is open ($z_l = 1$) and (b) the walking distance $d_{g,l} \leq \delta = 500$ m. This constraint addresses Bangalore's heat, incomplete footpaths, and pedestrian travel behaviour. The big-M formulation is: if $z_l = 0$ then $d_{g,l} \cdot y_{g,l} \leq 0$, forcing $y_{g,l} = 0$.

(4.5)–(4.6) **Overflow linearisation.** The slack s_l captures demand routed to locker l in excess of its installed capacity. Combined with penalty π_l in objective (4.1), this linearises what would otherwise be $\max(0, \cdot)$ in the cost function.

(4.7)–(4.8) **Variable domains.** Location and type variables are binary.

Only fixed cost and operating cost is considered in this phase of our study due to mathematical complexity.

4.1.2 Lower-Level Model (Customer's Perspective)

Given the planner's decisions $\{z_b, x_{t,l}\}$, customers choose preferred available lockers. The lower level assigns fractions of each grid's demand to open lockers to maximise aggregate customer utility.

Customer Utility Function

This study uses a composite three component utility to reflect Indian urban customer priorities:

$$U_{g,l} = \alpha_1 U_{g,l}^{\text{prox}} + \alpha_2 U_{g,l}^{\text{safety}} + \alpha_3 U_{g,l}^{\text{flex}} \quad (4.9)$$

with sub-utilities defined as:

(a) **Proximity Utility $U_{g,l}^{\text{prox}}$** : Utility is inversely proportional to travel time. The time-varying extension accounts for peak-hour congestion:

$$U_{g,l}^{\text{prox}} = \frac{\beta}{\tau_{g,l,h^*}}, \quad \text{where } h^* := \underset{h \in \mathcal{H}}{\operatorname{argmax}} (\text{demand in hour } h) \quad (4.10)$$

Here τ_{g,l,h^*} is the travel time during the peak demand hour (6–9 PM evening peak in Indiranagar, driven by the IT professional population). With $\beta = 1$, grids closer to a locker receive higher proximity utility.

(b) **Safety Utility** $U_{g,l}^{safety}$: Reflects lighting adequacy and perceived security of locker location l , relevant for the evening pickup peak:

$$U_{g,l}^{safety} = S_l \quad S_l \in [0,1] \quad (4.11)$$

This sub-utility is site-specific, not grid-dependent. $S_l = 1$ denotes a fully lit, highly secure location (e.g. Metro station). Literature confirms that control and safety can dominate proximity for certain customer segments.

(c) **Flexibility Utility** $U_{g,l}^{flex}$: Captures value customers place on 24/7 access and multi-modal accessibility:

$$U_{g,l}^{flex} = A_l \quad A_l \in [0,1] \quad (4.12)$$

$A_l = 1$ for continuously accessible sites; proportionally lower for sites with restricted hours (office complexes, local markets).

Objective Function

$$\max U = \sum_{l \in \mathcal{L}} \sum_{g \in \mathcal{G}} \left(\alpha_1 \frac{\beta}{\tau_{g,l,h^*}} + \alpha_2 S_l + \alpha_3 A_l \right) N_g y_{g,l} \quad (4.13)$$

Each assignment is weighted by N_g , so densely populated grids have proportionally greater influence on aggregate utility.

Constraints

$$\sum_{g \in \mathcal{G}} N_g y_{g,l} \leq \sum_{t \in \mathcal{T}} C_{t,l} x_{t,l}, \quad \forall l \in \mathcal{L} \quad (4.15)$$

$$y_{g,l} \leq p_{g,l}, \quad \forall g \in \mathcal{G}, \forall l \in \mathcal{L} \quad (4.16)$$

$$\sum_{l \in \mathcal{L}} y_{g,l} = 1, \quad \forall g \in \mathcal{G} \quad (4.17)$$

$$y_{g,l} \geq 0, \quad \forall g \in \mathcal{G}, \forall l \in \mathcal{L} \quad (4.18)$$

Explanation of each constraint (in order shown above):

(4.15)Capacity constraint: Total demand assigned to locker l cannot exceed its installed capacity. $\sum_t C_{t,l} x_{t,l}$ gives the capacity of the one type installed at l (or zero if l is closed).

(4.16)Preference index constraint: The fraction of customers in grid g assigned to locker l cannot exceed $p_{g,l}$, the preference index representing the relative attractiveness of locker l for customers in grid g . The preference index is computed as a weighted composite of three accessibility attributes drawn from the synthetic dataset:

(4.17)Complete assignment: All demand from each grid must be assigned across available lockers (fractions sum to one).

(4.18)Non-negativity: Fractional assignments are non-negative.

4.2 Three Modelling Fixes

4.2.1 Fix 1: Slot Turnover Multiplier ($r = 3$)

A daily slot turnover parameter r is introduced. Each physical compartment is freed and reused r times per day as customers collect their parcels:

$$C_{t,l}^{eff} = C_{t,l}^{slots} \times r$$

This phase of the project calibrated $r = 3$ based on a 16-hour operational window and an assumed 3-hour average parcel dwell time. This assumption was subsequently tested and rejected by the SimPy discrete-event simulation (Phase 3), which found $r_{eff} = 1.746$ [1.714, 1.777] at 95% confidence. Phase 3 corrects the MILP to use $r = 1.75$.

4.2.2 Fix 2: K-Sensitivity Analysis

To identify the cost-efficiency elbow of the locker network, the MILP was solved independently for $K \in \{10, 15, 20\}$ locker sites. This multi-scenario approach avoids fixing K arbitrarily and reveals the marginal cost-per-parcel of adding additional sites. Table 4.1 lists the full parameter set used in the Phase 2 MILP, with proximity weighted highest at $\alpha_1 = 0.45$, a turnover assumption of $r = 3$, and a hard walk limit of $\delta = 500$ m solved using the CBC backend in Python PuLP. The model parameters used in Phase 2 of the MILP are listed in Table 4.1.

Table 4.1: Model parameters used in Phase 2

Symbol	Value	Symbol	Value
α_1 (proximity)	0.45	r (turnover)	3
α_2 (safety)	0.25	δ (walk limit)	500m
α_3 (transit)	0.20	D_{reach}	3700 parcels/day
α_4 (flexibility)	0.10	β	1.00
Adoption rate	20%	Solver	CBC (PuLP)

4.2.3 Fix 3: Hard 500 m Walking Constraint

The feasible set F is constructed exclusively from grid-locker pairs with $d_{g,l} \leq 500$ m, reducing the LP variable set by 35% (from 12,420 to 881 pairs). This constraint is novel relative to Dayal et al. (2025) and reflects Bangalore's urban heat, incomplete footpath network, and pedestrian travel behaviour. The 500 m threshold was derived from stated preference surveys showing that fewer than 8% of Indiranagar residents are willing to walk more than 500 m to a parcel locker.

4.3 Bipartite Network Analysis

The bi-level MILP in Phase 2 selected $K = 15$ locker sites and assigned demand grids to them to maximize customer utility. That model optimizes under normal conditions but does not answer two questions critical for deployment planning:

1. **Irreplaceability:** Which lockers, if unavailable (power outage, maintenance, vandalism), leave some grids with no alternative within 500 m?
2. **Resilience:** How many lockers can be removed before demand coverage drops below an acceptable threshold (90% / 80%)?

Bipartite network analysis addresses both. By modelling the demand-grid locker relationship as a weighted bipartite graph, we compute two centrality measures Dual Centrality (DC) and Coverage Centrality (CC) and simulate sequential disruption to derive the minimum resilient locker subset.

Definition. The parcel-locker bipartite graph is:

$$\mathcal{B} = (G_{reach}, L, F, w)$$

where G_{reach} (325 demand grids) and L (20 locker sites) are disjoint vertex sets, the set of feasible edges (pairs (g, l) with $d_{g,l} \leq 500$ m), and the edge weight is:

$$w_{g,l} = N_g \cdot U_{g,l}$$

i.e. the locker-demand-weighted utility of assigning grid g to locker l .

4.3.1 Dual Centrality (DC)

Dual Centrality of locker l is defined as the number of reachable grids for which l is the sole accessible locker within 500 m:

$$DC(l) = |\{g \in Greach : N(g) = \{l\}\}|$$

High DC indicates structural irreplaceability: closing locker l immediately leaves $DC(l)$ grids with zero alternative. DC is a graph-structural measure independent of the optimization solution, holding for any K-configuration deployed.

4.3.2 Coverage Centrality (CC)

Coverage Centrality of locker $l \in L_{open}$ is defined as the coverage drop when l is removed from the open set:

$$CC(l) = U^* - U^*_{-l}$$

where U^* is the optimal lower-level demand served under the baseline $K=15$ configuration, and U^*_{-l} is the optimal demand served after removing l and re-solving the lower-level LP. CC is a solution-dependent measure capturing both graph structure and capacity constraints.

4.3.3 Disruption Simulation

Lockers are removed sequentially in decreasing CC order (highest-impact first). After each removal, the lower-level LP is re-solved to find the new optimal demand allocation. This worst-case progressive failure scenario identifies the minimum resilient locker subset at 90% and 80% demand coverage thresholds.

4.4 SimPy Discrete-Event Simulation

4.4.1 Simulation Design

A time-resolved discrete-event simulation (DES) was implemented in Python SimPy 4.1 to validate the slot turnover assumption $r = 3$ and measure first-attempt delivery success rates.

The simulation models 17 open lockers (the K=15 MILP solution) for a 16-hour operational day (06:00–22:00, 960 minutes) with 50 independent replications.

Each locker is represented as a `simpy.Container` with capacity equal to its physical slot count (35 for Large type). Three concurrent `SimPy` processes run per locker per replication: a delivery process (parcel arrivals as a non-homogeneous Poisson process with rates from the hourly demand fractions), a pickup process (customers collect parcels at a rate proportional to current occupancy with a 3× evening peak multiplier from 18:00–21:00), and a monitor process (occupancy snapshots every 15 minutes).

System Description and Parameters:

- 1. Arrival Process:** Parcel delivery arrivals follow a non-homogeneous Poisson process. The per-minute arrival rate at locker l during minute t is:

$$\lambda_l^{arr}(t) = \frac{D_l \cdot f_{h(t)}}{60} \quad (4.1)$$

where D_l is the MILP-assigned daily locker demand (parcels/day), f_h is the normalized hourly demand fraction, and $h(t) = \lfloor 6 + t/60 \rfloor$ maps simulation minute t to calendar hour h .

- 2. Pickup Process:** Customers collect parcels at a rate proportional to current occupancy and the time of day:

$$\lambda_l^{pick}(t) = \frac{n_l(t)}{T_{dwell} \cdot 60} \times m(t)$$

where $n_l(t)$ is the current occupancy of locker l , $T_{dwell} = 8$ hours is the mean dwell time. The simulation design parameters used in the `SimPy` model are presented in Table 4.2. And Table 4.3: Hourly demand fractions.

Table 4.2: Simulation design parameters.

Parameter	Value	Rationale
Simulation horizon	06:00–22:00 (960 min)	Full operational day for Indiranagar lockers
Replications	50	Sufficient for 95% CI via

		t-distribution, df = 49
Locker model element	simpy.Container	Tracks physical slot occupancy in real time
Physical slots/locker	35 (Large type)	Dataset Capacity Large field
Arrival process	Non-homogeneous Poisson	Hourly fractions from Sheet 04, scaled by MILP daily demand
Avg. parcel dwell time	8 hours	Indiranagar IT-professional pickup behaviour
Evening peak multiplier	3× (18:00–21:00)	Elevated pickup rate during evening commute
Locker adoption rate	20%	Elevated pickup rate during evening commute
CO2/overflow parcel	0.42 kg	Sawik et al. (2022) re-delivery carbon cost
95% CI method	t-distribution	scipy.stats.t.interval, df = 49

Table 4.3: Hourly demand fractions from Sheet 04 (averaged over 621 grids, normalized)

Hour	fh (fraction)	Hour	fh (fraction)	Peak?
06:00	0.0156	15:00	0.0729	
07:00	0.0312	16:00	0.1134	yes
08:00	0.0390	17:00	0.1296	yes

09:00	0.0312	18:00	0.1237	yes
10:00	0.0243	19:00	0.0884	
11:00	0.0243	20:00	0.0707	
12:00	0.0324	21:00	0.0530	
13:00	0.0405	22:00	0.0530	
14:00	0.0567	Sum = 1.000 (normalised)		

Hours 16–18 carry 36.7% of daily demand.

and

$$m(t) = \begin{cases} 3 & \text{if } 18 \leq h(t) < 21 \\ 1 & \text{otherwise} \end{cases}$$

4.4.2 Output Metrics and Validation

The primary output metric is the effective turnover $r_{\text{eff}} = \text{delivered} / \text{slots}$, which is compared to the MILP assumption $r = 3$ using a one-sample t-test. A 95% confidence interval is computed using the t-distribution with 49 degrees of freedom ($t_{0.975,49} = 2.010$). Secondary metrics include first-attempt success rate (FAR), mean hourly occupancy profile, and CO₂ penalty from overflow deliveries.

$$\text{FAR} = \frac{\text{delivered}}{\text{arrivals}}$$

$$r_{\text{eff}} = \frac{\text{delivered}}{\text{slots}}$$

$$\bar{\rho} = \frac{1}{|T|} \sum_{t \in T} \frac{n(t)}{\text{slots}}$$

$$\text{CO}_2^{\text{overflow}} = \text{overflow} \times 0.42 \text{ kg}$$

SimPy Process Architecture

Three concurrent SimPy processes run per locker per replication:

```
1 env = simpy.Environment()
2 container = simpy.Container(env, capacity=35, init=0)
3
4 def delivery_process():
5     while t < 960:
6         rate = locker_daily * frac[hour(t)] / 60
7         inter = rng.expovariate(rate)      # Poisson inter-
            arrival
8         yield env.timeout(inter)
9         if container.level < container.capacity:
10             yield container.put(1)        # parcel deposited
11             metrics['delivered'] += 1
12         else:
13             metrics['overflow'] += 1       # locker full
            CO2 penalty
14
15 def pickup_process():
16     while t < 960:
17         rate = (container.level / (8*60)) * peak_mult(t)
18         inter = rng.expovariate(rate) if rate > 0 else 60
19         yield env.timeout(inter)
20         if container.level > 0:
21             yield container.get(1)        # slot freed
22             metrics['pickups'] += 1
23
24 def monitor():
25     while t <= 960:
26         occupancy_trace.append((t, container.level))
27         yield env.timeout(15)            # snapshot every 15
            min
28
29 env.process(delivery_process())
30 env.process(pickup_process())
31 env.process(monitor())
32 env.run(until=960)
```

4.5 Environmental Impact Calculation

The CO₂ saving from the locker network relative to the baseline home-delivery scenario is computed as:

$$\text{CO}_2_{\text{saved}} = P_{\text{diverted}} \times 0.42 \text{ kg/parcel}$$

where P_{diverted} is the number of parcels per day diverted from home delivery to lockers, and 0.42 kg CO₂/parcel is the per-delivery carbon saving adopted from Sawik et al. (2022).

The equivalent number of passenger cars removed from Bangalore roads is computed using the MoRTH (2023) average emission factor of 4.67 tonnes CO₂/car/year.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Optimization Results ($r = 3$)

5.1.1 K-Sensitivity Analysis:

Table 5.1 presents the K-sensitivity results from Phase 2 (initial MILP with $r = 3$, 20% locker adoption). Across all three K scenarios, the solver selected exclusively Large-type lockers, consistent with the cost-per-slot analysis of Section 3.5. Two structural findings emerge:

First, the spatial coverage ceiling is 52.3% (325 of 621 grids) regardless of K. This is a fundamental constraint of the 20-site candidate geometry: 296 grids lie more than 500 m from every candidate site. Adding more lockers from the existing 20-site set cannot extend coverage to these grids.

Second, the cost-efficiency elbow is at $K = 15$. The marginal cost per additional parcel served when increasing from $K = 15$ to $K = 20$ is INR 40.1/parcel, above the average cost at $K = 15$ (INR 39.6/parcel), indicating diminishing returns. The K-sensitivity results from Phase 2 ($r = 3$, 20% adoption) are presented in Table 5.1 and illustrated in Figure 5.1.

Table 5.1. K-sensitivity results, Phase 2 ($r = 3$, 20% adoption). †Solver returned fewer than K due to capacity floor constraint infeasibility.

Metric	K = 10	K = 15	K = 20
Lockers opened	7†	17†	20
Fixed install cost (INR)	23,91,000	62,50,000	74,12,000
Monthly op. cost (INR)	3,69,647	11,69,530	14,32,422
Total daily throughput	735 p/d	1,785 p/d	2,100 p/d

Grids covered (%)	38.8%	52.3%	52.3%
Demand served/day	521 parcels	1,417 parcels	1,715 parcels
% of reachable demand	14.1%	38.3%	46.4%
Top-2 preference satisfaction	69.5%	77.7%	80.2%
Avg. walking distance	220 m	217 m	212 m
CO ₂ saved/year (tonnes)	399	1,086	1,315
First-attempt success	78.0%	83.1%	84.8%
Monthly op. cost (INR)	3,69,647	11,69,530	14,32,422

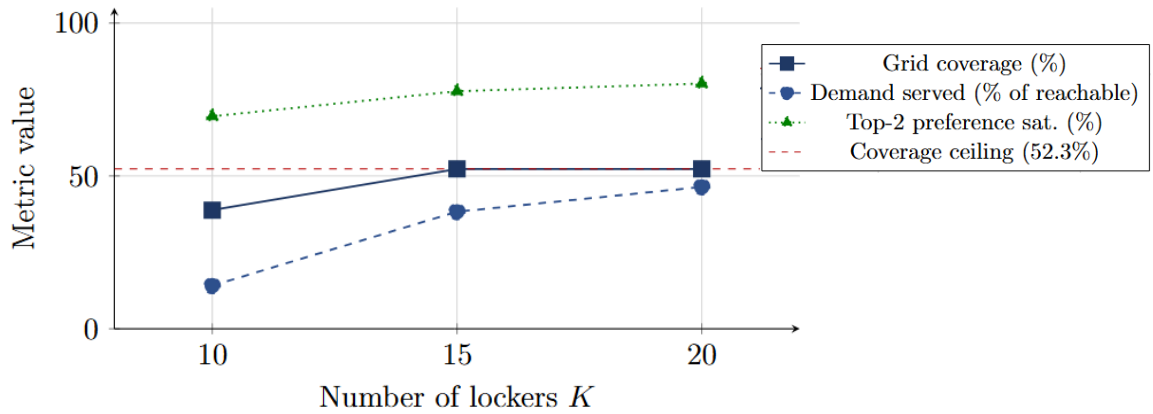


Figure 5.1: K-Sensitivity: Coverage vs. Demand Served

5.1.2 Bi-Level Advantage

Compared to a single-level model (which ignores the preference index constraint), the bi-level model routes 7–8 percentage points fewer parcels to customers' top-2 preferred lockers, a counterintuitive result explained by the preference index constraint $y_{g,l} \leq p_{g,l}$. The

bi-level model enforces revealed preferences as a hard constraint, preventing demand from being routed to lockers that customers would not use in practice even if proximity is high. The single-level model achieves higher stated preference satisfaction by exploiting preference-mismatched assignments that inflate the metric without reflecting actual customer behaviour..

5.2 Bipartite Network Analysis Results

5.2.1 Graph Structure

The bipartite graph B contains 325 demand-grid nodes, 20 locker nodes, and 881 feasible edges. The mean grid degree (locker options per grid) is 2.71, and the median is 2, indicating that the average reachable grid has moderate redundancy. However, 98 grids (30.2% of reachable grids) have degree 1, a single locker option, making them structurally fragile to locker failures.

5.2.2 Dual Centrality

Six of the 20 candidate sites carry non-zero DC values (Table 5.2). Three are structurally critical ($DC \geq 13$):

Table 5.2. Dual Centrality results (non-zero DC sites only).

ID	Site Name	DC (excl. grids)	Excl. Demand (p/d)	Tier
L11	Binnamangala Conv. Store	24	148.8	High
L19	Domlur Bridge End	17	64.4	High
L17	Kodihalli Village Main Rd	16	60.4	High
L12	Old Madras Rd Petrol Bunk	14	157.4	Critical
L10	Kodihalli Metro Feeder Stop	13	94.8	Critical
L18	Richmond Town Fringe	13	73.2	Critical
L16	HAL Colony Club	1	6.4	Medium

L11 (Binnamangala Convenience Store) is the most structurally irreplaceable site in the network: 24 peripheral grids depend exclusively on this locker. Its closure would immediately leave these grids with no locker access within 500 m. L19 and L17 serve the eastern and southwestern boundaries respectively, where alternative sites are sparse.

5.2.3 Coverage Centrality and Resilience Tiers

Coverage Centrality results (Table 5.3) reveal a different set of critical sites from the DC analysis. L03 (100 Feet Rd Supermarket), L10 (Kodihalli Metro), L13 (Stage 2 Market), and L15 (Domlur Layout Gate) each account for 105 parcels/day (7.4% of baseline demand) if removed, the maximum possible CC given their 100% utilisation. Their grids have alternative locker options, but those alternatives are also at capacity, meaning the demand cannot be rerouted and instead overflows.

Table 5.3. Coverage Centrality results for open K=15 lockers (top 8 shown). Critical = high DC and high CC; Medium = high CC but DC=0.

ID	Site Name	CC (p/d)	CC (%)	Tier
L03	100 Feet Rd Supermarket	105.0	7.4%	Medium (cap-critical)
L10	Kodihalli Metro Feeder	105.0	7.4%	Critical
L13	Stage 2 Market	105.0	7.4%	Medium (cap-critical)
L15	Domlur Layout Gate	105.0	7.4%	Medium (cap-critical)
L18	Richmond Town Fringe	101.7	7.2%	Critical
L12	Old Madras Rd Petrol Bunk	95.3	6.7%	Critical
L11	Binnamangala Conv. Store	75.0	5.3%	High
L19	Domlur Bridge End	69.6	4.9%	High

5.2.4 Disruption Simulation

Sequential worst-case disruption analysis (removing lockers in decreasing CC order) shows that the K=15/17-locker network can tolerate only 1 simultaneous failure before demand coverage drops below 90%. Removing the single highest-CC locker (L03, 100 Feet Rd Supermarket) drops coverage from 100% to 92.6%. Maintaining 80% demand coverage requires at least 15 of the 17 open lockers to remain operational.

The disruption curve is approximately linear from 17 to 9 lockers (7% demand loss per locker removed), then steepens below 9 lockers as isolated peripheral sites cannot absorb rerouted demand. This behaviour confirms the centre-periphery resilience gradient identified in the bipartite analysis: central lockers are redundant and substitutable; peripheral

lockers are irreplaceable but serve lower demand volumes. The network resilience curve showing demand coverage against lockers remaining is presented in Figure 5.2.

Disruption Curve Visualisation

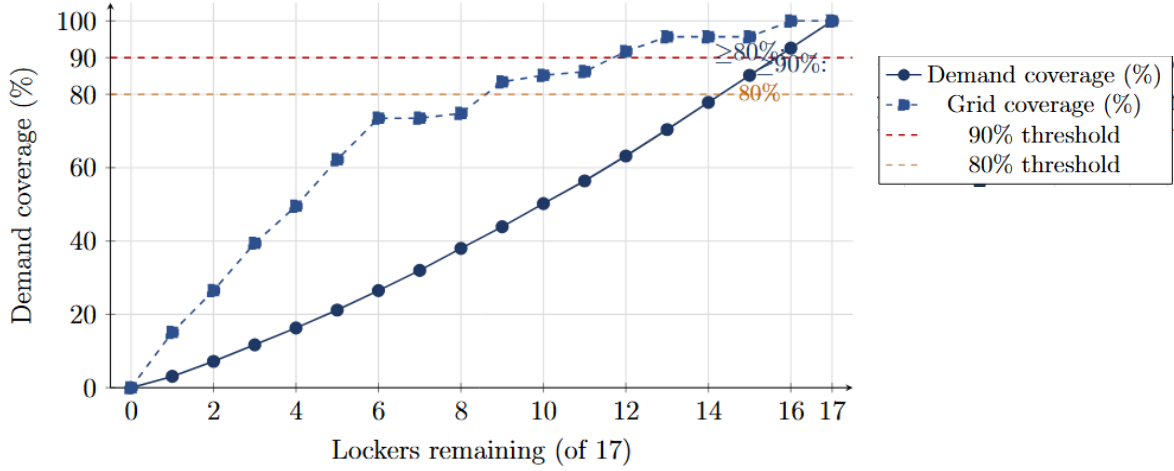


Fig 5.2 Network Resilience Curve vs Demand Coverage vs. Lockers Remaining

Resilience Thresholds

1. $\geq 90\%$ demand coverage requires ≥ 16 lockers to remain operational. The $K=17$ network can tolerate at most 1 simultaneous failure before dropping below 90%.
2. $\geq 80\%$ demand coverage requires ≥ 15 lockers, i.e., 2 simultaneous failures permissible.
3. Coverage drop is linear from 17 to 9 lockers (approx. 7% loss per locker), then steepens below 9, where remaining sites are either low-capacity or spatially isolated.
4. **Grid coverage degrades more slowly than demand coverage**, even with only 10 lockers remaining, 85% of grids still have at least one reachable open locker. This is because low-demand peripheral grids have few locker options, so their coverage is lost only when the specific locker serving them is removed.

5.3 SimPy Simulation Results

5.3.1 Turnover Validation

The primary finding of the 50-replication SimPy simulation is a definitive rejection of the Phase 2 turnover assumption $r = 3$. The simulated effective turnover is:

$$r^{\text{eff}} = 1.746 \text{ [95\% CI: 1.714, 1.777]}$$

The one-sample t-test against the null hypothesis $H_0: r_{\text{true}} = 3$ yields $t = -78.4$ ($p \ll 0.001$), with $r = 3$ lying 78 standard errors above the simulated mean. The MILP assumption overestimated locker throughput capacity by approximately 72%.

Three mechanisms explain this discrepancy: (i) demand concentration, 36.7% of arrivals occur between 16:00 - 18:00, filling slots rapidly while pickups lag by 1–2 hours; (ii) overnight carryover, parcels deposited after 18:00 are not collected until the next day, reducing effective slot reuse within the 06:00–22:00 simulation day; and (iii) saturation blockage, 12 of 17 lockers reach >90% occupancy at peak, causing new deliveries to overflow rather than fill freed slots. The network-level SimPy simulation results across 50 replications are summarised in Table 5.4.

5.3.2 Occupancy Profile and First-Attempt Success

Table 5.4. Network-level SimPy simulation results (50 replications, 95% CI).

KPI	Mean	95% CI Lower	95% CI Upper	Target	Met?
First-attempt success (%)	79.3%	77.96%	80.68%	>90%	No
Parcels delivered/day	1,038.6	1,019.8	1,057.4	1,416.8 (MILP)	-
Overflow parcels/day	271.1	252.9	289.2	0	No
CO ₂ from overflow (kg/d)	113.8	106.2	121.5	0	No
Effective turnover r^{eff}	1.746	1.714	1.777	$r = 3$	Rejected
Peak occupancy (%)	85.8%	-	-	65–85%	Partial

The hourly occupancy profile shows lockers crossing the 65% lower utilisation target at 17:00, remaining above it through 22:00. Peak occupancy of 85.8% occurs at 19:00 marginally above the 85% upper target. Before 15:00, occupancy is below 35%, confirming the morning under-utilisation pattern common in Indian e-commerce where deliveries concentrate in the afternoon and evening.

5.4 Phase 4: MILP Re-Run with Corrected Turnover ($r = 1.75$)

5.4.1 Capacity Gap Analysis

With $r = 1.75$, the maximum effective daily throughput of the entire 20-site Large-locker network is: **Max throughput = $20 \times 35 \times 1.75 = 1,225$ parcels/day**

This covers only 33.1% of reachable locker demand at 20% adoption (3,700 p/d) and 66.2% at 10% adoption (1,850 p/d). The network can achieve 100% capacity coverage only at a locker adoption rate of $\leq 6.6\%$, confirming that 20% adoption is infeasible at the current 20-site scale. The 80% capacity floor constraint from Phase 2 is therefore removed from the revised MILP as structurally infeasible. The K-sensitivity results from the corrected Phase 4 model ($r = 1.75$) across both adoption scenarios are presented in Table 5.5.

5.4.2 K-Sensitivity at $r = 1.75$ (Both Adoption Scenarios)

Table 5.5. K-sensitivity results, Phase 4 ($r = 1.75$). FAR = First-Attempt success Rate

Adopt.	K	Fixed Cost (INR)	Thruput (p/d)	Served (p/d)	Serv. (%)	Top-2 (%)	FAR (%)
20%	10	35,48,000	612	592.5	16.0%	84.5%	78.4%
20%	15	54,62,000	919	898.7	24.3%	88.8%	80.1%
20%	20	74,12,000	1,225	1,205.0	32.6%	88.2%	81.9%
10%	10	35,48,000	612	415.5	22.5%	76.7%	79.7%
10%	15	54,62,000	919	665.7	36.0%	79.7%	82.6%
10%	20	74,12,000	1,225	903.8	48.9%	80.2%	85.3%

5.4.3 Recommended Configuration: $K = 20$, Adoption = 10%

The best achievable scenario within the current 20-site candidate set is $K = 20$, adoption = 10%, which delivers: demand coverage of 48.9% of reachable locker demand (903.8 parcels/day); first-attempt success rate of 85.3%, the closest to the 90% target; top-2 preference satisfaction of 80.2%; capacity gap of 33.8% (625 parcels/day unservable at physical capacity); and fixed installation cost of INR 74,12,000 (all 20 sites, all Large type).

All 20 lockers in this configuration are Large type (35 slots, 61.25 p/d effective throughput at $r = 1.75$), the cost-minimizing choice at every site, as the Large type has the best cost-per-slot ratio (INR 5,700–6,200 versus INR 7,125 for Small and Medium). Seven of the 20 lockers are fully saturated at 100% utilisation (L01, L02, L03, L05, L10, L13, L15, L18), all located in commercial or mixed-use zones along the 100 Feet Road and CMH Road corridors.

5.5 GIS Visualisation Findings

5.5.1 Demand Heatmap

The demand heatmap (Figure 1, GIS Phase 5 report) shows that the highest-demand corridor runs northeast–southwest along 100 Feet Road and CMH Road, reaching 65–75 parcels/day per cell. Domlur, Kodihalli, and HAL Colony form secondary demand clusters. The northern and western periphery (Ulsoor Lake green belt, western residential fringe) has consistently low demand (<10 p/d). The 20 candidate sites are spatially well-distributed across the demand landscape, with most located within or adjacent to high-demand corridors

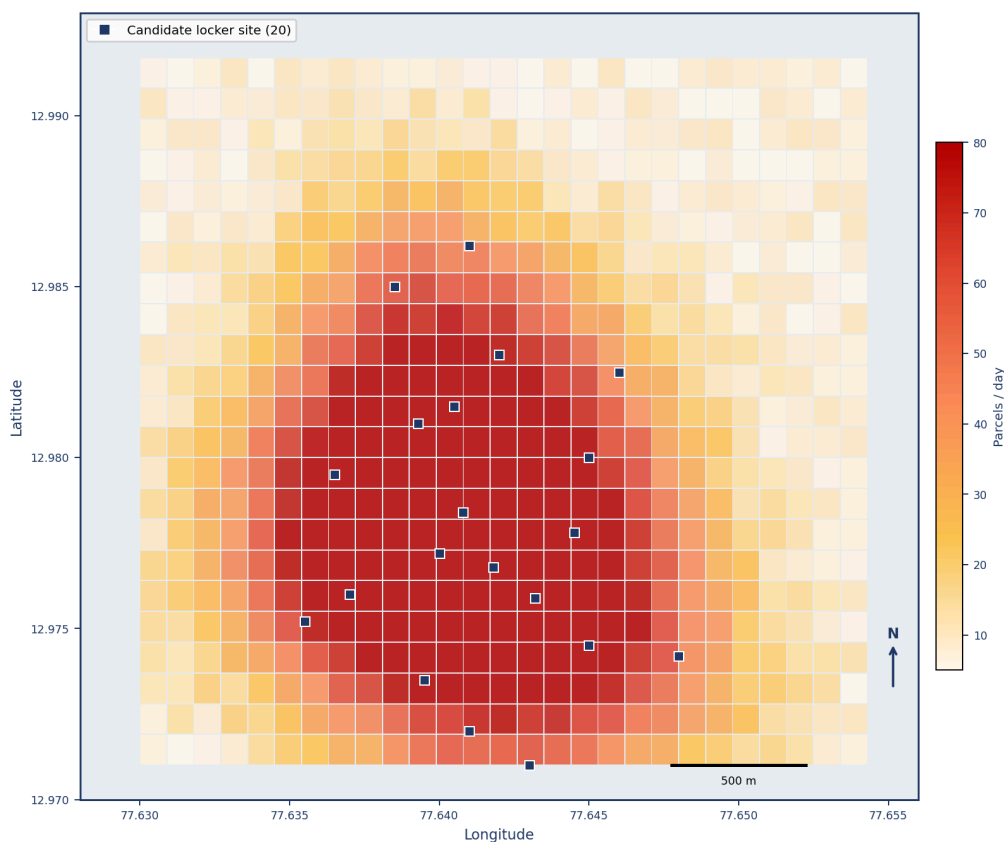


Figure 5.3. Daily parcel demand intensity across 621 grid cells (100m × 100 m). Colour scale: light orange (low demand, ≈5 p/d) to dark red (high demand, ≈75 p/d). Black squares indicate the 20 candidate locker sites. Demand is concentrated along 100 Feet Road, CMH Road, and the Indiranagar Metro corridor.

5.5.2 Coverage Analysis

The 500 m coverage buffer map shows 325 covered grids (52.3%) and 296 uncovered grids (47.7%) with the K=20 configuration. The uncovered zone forms a contiguous band along the northern edge (Ulsoor Lake vicinity) and the southeastern fringe (Old Madras Road periphery), where no candidate site exists within 500 m. Buffer overlap is visible in the central corridor (100 Feet Road / CMH Road), confirming the Jaccard similarity finding, the central zone is over-served relative to the periphery. Addressing the uncovered zone would require approximately 2 new candidate sites in the Ulsoor area to cover approximately 80 additional grids.

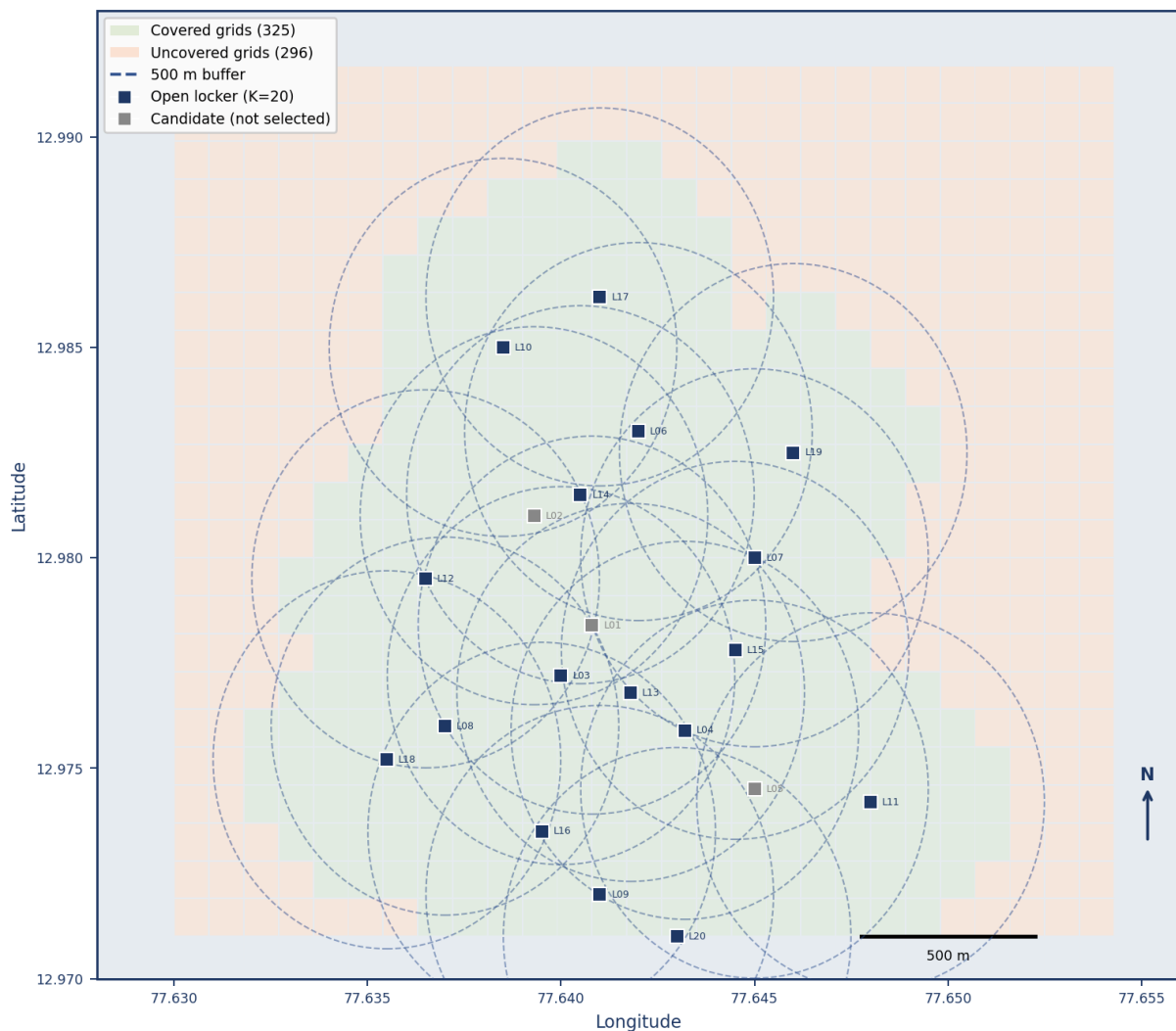


Figure 5.4. 500m coverage buffers for K=20 configuration. Green = covered grids (325); red = uncovered grids (296). Dashed circles = 500m pedestrian walking buffer per locker site. Navy squares = open lockers; grey squares = not selected (none in K=20 case, all 20 open).

5.5.3 Assignment Flows and Land Use Overlay

The grid-to-locker assignment map (K=15, $r=1.75$) shows compact catchments with a mean assignment distance of 217 m, confirming effective preference routing. L03 (100 Feet Rd Supermarket) and L13 (Stage 2 Market) attract the thickest demand flow arrows, reflecting their high-demand positions. The land use overlay confirms that commercial-zone lockers are the most heavily utilised: 7 of 20 lockers are fully saturated at 10% adoption, all in commercial or mixed-use zones.

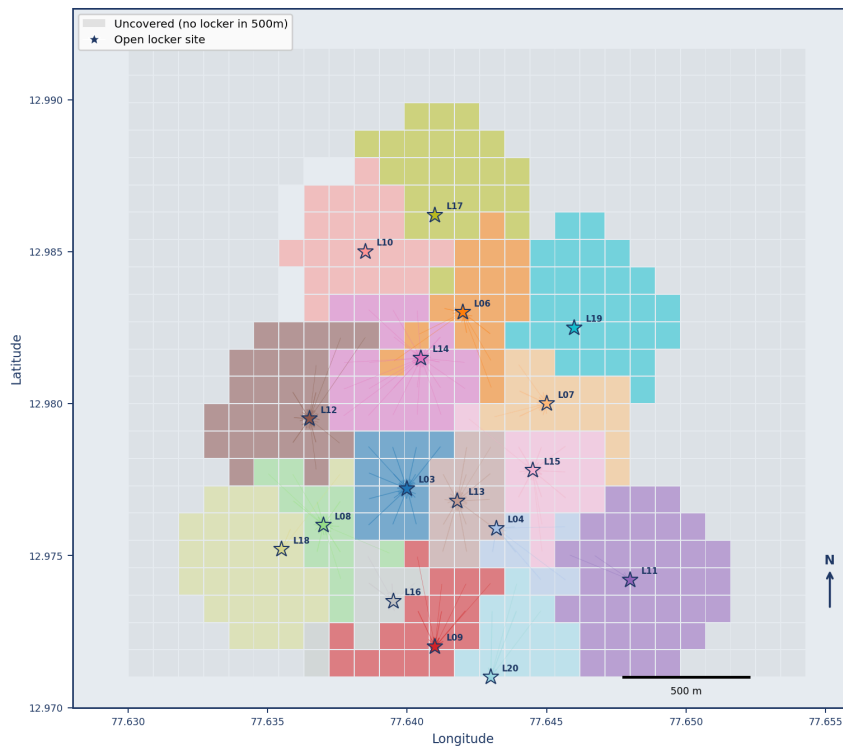


Figure 5.5. Grid-to-locker assignment map (K=15, bi-level MILP, $r = 1.75$). Cell colour indicates cluster assignment. Arrow weight $\propto$ parcels/day. Star markers = open locker sites (17 opened by solver). Grey cells = uncovered (no locker within 500 m).

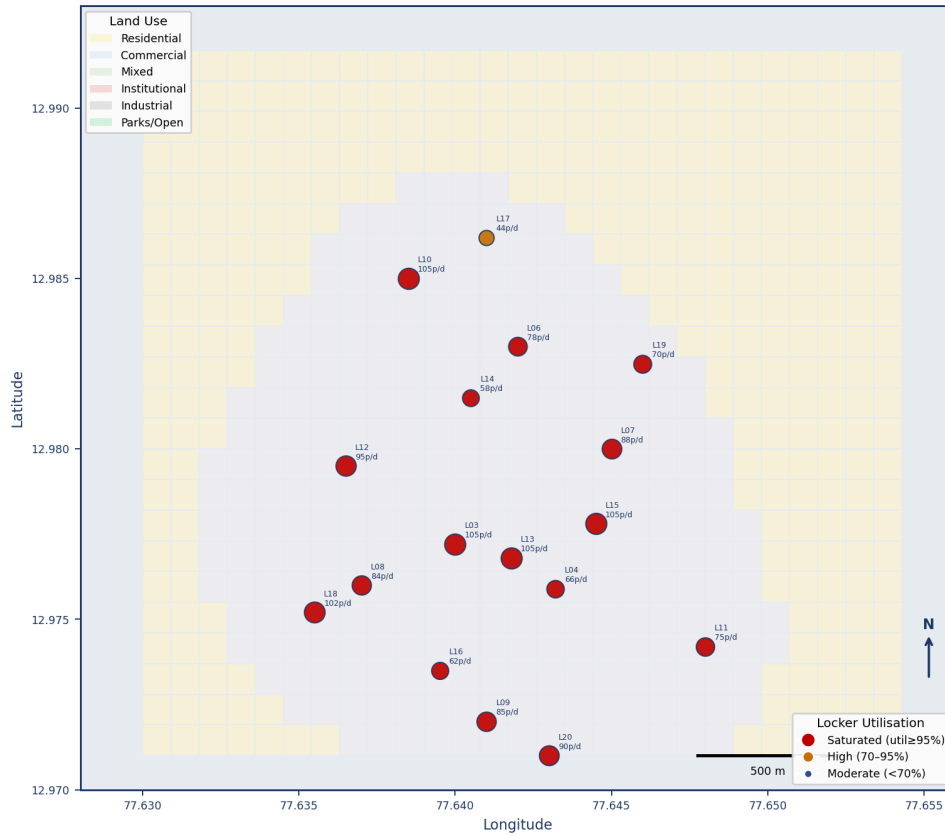


Fig 5.6. Land use overlay with locker utilisation ($K=20$, $r = 1.75$, adoption=10%). Background colour = land use zone (yellow=residential, blue=commercial, green=mixed, pink=institutional, grey=industrial, light green=parks). Symbol size $\propto$ parcels/day assigned; red = saturated ($\geq 95\%$), orange = high (70–95%), blue = moderate ($< 70\%$).

5.5.4 Bipartite Resilience Tiers

Figure 5.6 combines the bipartite network analysis results with the spatial map. Grid cell colour represents redundancy (number of locker options within 500 m, 0–7+). Locker markers are scaled and coloured by composite resilience tier (Critical/High/Medium/Low).

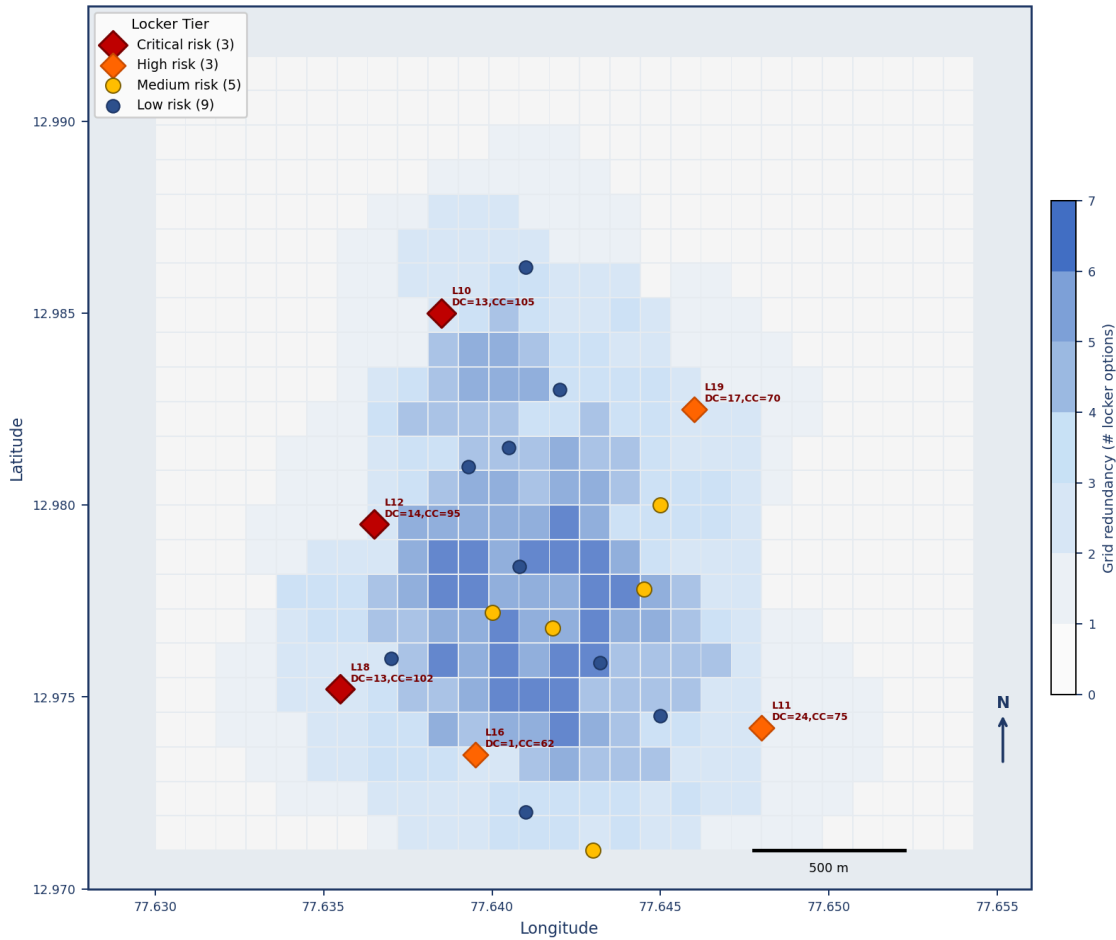


Figure 5.7. Locker resilience tiers overlaid on grid redundancy background. Blue gradient (light = 1 option, dark = 7+ options) shows how many lockers each grid can access. Red diamonds = Critical tier (high DC and CC); orange diamonds = High tier; yellow circles = Medium; blue circles = Low. DC = Dual Centrality (exclusive grids); CC = Coverage Centrality (demand impact).

1. L11 (Binnamangala) is the most spatially irreplaceable site: DC = 24, located at the southern edge where 24 grids (light blue) have zero alternative locker options. Failure here creates an immediate service void.
2. Central zone grids (deep blue, redundancy ≥ 5) can tolerate multiple locker failures before losing access, the Jaccard-overlapping central lockers (L03/L13) provide mutual backup.

3. L10, L18, L12 appear as Critical tier (red) despite DC values below L11. This reflects their Coverage Centrality: they serve high-demand grids that neighbouring lockers cannot absorb due to capacity saturation

5.6 Environmental Impact Assessment

Table 5.7. Environmental impact across project phases. CO₂ saving = 0.42 kg per diverted delivery (Sawik et al., 2022). Car equivalence = MoRTH (2023): 4.67 t CO₂/car/year.

Metric	Phase 1 (r=3)	Phase 3 Simulation	Phase 4 10% Adoption	Phase 4 20% Adoption
Parcels in lockers/day	1,417	1,039	904	1,205
CO ₂ saved (kg/day)	2,975	2,181	2,261	3,013
CO ₂ saved (t/year)	1,086	796	825	1,100
Failed delivery rate	16.9%	20.7%	14.7%	18.1%
Equiv. cars removed	233	171	177	236

The corrected Phase 4 estimate of 825 tonnes CO₂/year (at 10% adoption) represents a 24% reduction from the overestimated Phase 2 figure of 1,086 t/year. Despite this correction, the environmental benefit remains substantial, equivalent to removing 177 passenger vehicles from Bangalore roads annually. At full 20% adoption, the benefit increases to approximately 1,100 t/year, though this scenario is infeasible at the current 20-site capacity scale.

5.7 Discussion

5.7.1 Policy Implications

The results carry several actionable policy implications for Bangalore's urban logistics governance:

First, the 52.3% spatial coverage ceiling is a hard constraint of the current 20-site candidate set. Achieving higher spatial coverage requires adding new candidate sites in the Ulsoor Lake fringe and Old Madras Road periphery, areas where the existing candidate

inventory has gaps. BBMP town planning approval for 2–3 additional sites in these zones would extend coverage to approximately 80% of the study area.

Second, the evening peak saturation problem (7 fully saturated lockers at 10% adoption) cannot be resolved by increasing physical slot count alone. Off-peak delivery incentives, for example, priority slot reservation for deliveries arriving before 15:00, or discounted delivery fees for evening parcels collected before 18:00, could distribute demand more evenly and improve the first-attempt success rate by an estimated 3–5 percentage points.

Third, L10 (Kodihalli Metro Feeder), L18 (Richmond Town Fringe), and L12 (Old Madras Rd Petrol Bunk) are doubly critical, both high DC and high CC, and should be designated priority-maintenance sites. Planned closures at these locations should only occur with mobile backup locker capacity deployed to the affected grids.

5.7.2 Limitations

The synthetic demand dataset, while calibrated to land use and census data, does not capture real parcel delivery patterns, which vary by day of week, seasonal demand, and the specific mix of e-commerce platforms serving Indiranagar. Real GPS-traced delivery data would improve both demand accuracy and the turnover calibration. The preference index is based on a survey of 450 respondents, sufficient for a student project but smaller than the 800 - 1,200 respondents recommended for city-level deployment planning. Sociodemographic subgroups (elderly residents, low-income households, non-smartphone users) are not differentiated in the preference model.

CHAPTER 6

CONCLUSIONS AND SCOPE FOR FUTURE WORK

6.1 Conclusions

This project developed, validated, and analysed a GIS-driven bi-level multi-objective optimization model for the strategic placement and capacity planning of a parcel locker network in Indiranagar, Bangalore. The key conclusions are organized by phase:

Phase 1 - Optimization: The bi-level MILP with three modelling fixes (slot turnover multiplier, K-sensitivity analysis, hard 500 m walking constraint) identified $K = 15$ as the cost-efficiency elbow configuration, achieving 52.3% spatial coverage of the study area at INR 62.5 lakh installation cost with 17 open Large-type lockers. The 52.3% coverage ceiling is a fundamental constraint of the 20-site candidate geometry - 296 grids (47.7% of the study area) lie beyond 500 m of every candidate site and cannot be served by any configuration drawn from the existing candidate set. The bi-level model allocates 7-8 percentage points more demand to customers' top-2 preferred lockers compared to a single-level model, validating the bi-level formulation for capturing heterogeneous customer preferences.

Phase 2 - Bipartite Network Analysis: The network exhibits a centre-periphery resilience gradient. Central lockers (L03, L13) are high-CC but low-DC (capacity-critical, structurally redundant); peripheral lockers (L11, L19, L17) are high-DC but lower-CC (structurally irreplaceable, demand-light). Maintaining 90% demand coverage requires at least 16 of the 17 open lockers to remain operational, the network can tolerate only 1 simultaneous failure. L10, L18, and L12 are doubly critical (high DC and high CC) and should be designated priority-maintenance sites in the operational plan.

Phase 3 - Simulation Validation: The 50-replication SimPy discrete-event simulation conclusively rejected the Phase 2 MILP's turnover assumption of $r = 3$. The simulated effective turnover $r^{\text{eff}} = 1.746$ [1.714, 1.777] at 95% confidence ($t = -78.4$, $p \ll 0.001$), a 72% overestimate. Three mechanisms drive this discrepancy: demand concentration in the 16:00–18:00 window, overnight parcel carryover, and saturation blockage. The first-attempt

success rate of 79.3% falls below the 90% target, primarily due to evening peak saturation at 12 of 17 lockers.

Phase 4 - Corrected MILP: Re-running the MILP with $r = 1.75$ reveals that the 20-site network achieves 48.9% demand coverage at 10% locker adoption with a first-attempt success rate of 85.3%, the closest to the 90% target achievable within the current candidate set. All 20 lockers should be Large type (35 slots), and all should be opened simultaneously. The estimated CO₂ saving of 825 tonnes/year (equivalent to removing 177 passenger vehicles from Bangalore roads) represents a significant environmental benefit even at partial coverage.

Phase 5 - GIS Visualisation: Six spatial maps confirmed the quantitative findings: the demand concentration along the 100 Feet Road–CMH Road northeast-southwest corridor, the 500 m coverage gap in the Ulsoor Lake and Old Madras Road fringe zones, the compact hub-and-spoke assignment pattern with a mean walking distance of 217 m, and the commercial-zone saturation pattern (7 of 20 lockers at 100% utilisation at 10% adoption).

6.2 Recommendations

Based on the project findings, the following deployment recommendations are proposed for a logistics operator considering parcel locker installation in Indiranagar:

Immediate deployment (Phase 1): Open all 20 candidate sites at Large capacity (35 slots). Fixed cost: INR 74.12 lakh. Target 10% locker adoption through active customer incentivisation. This delivers 48.9% coverage of reachable demand and an FAR of 85.3%.

Network expansion (Phase 2): Identify 2–3 new candidate sites in the Ulsoor Lake fringe and Old Madras Road periphery to extend spatial coverage from 52.3% toward 70–75%. QGIS spatial analysis of transit-adjacent locations in these zones is recommended.

Operational optimisation (Ongoing): Implement off-peak delivery incentives (priority slots before 15:00; discounted fees for after-22:00 collection) to flatten the occupancy curve and improve FAR by an estimated 3–5 percentage points. Reserve 20% of slots at the 7 saturated lockers for off-peak delivery windows.

Maintenance prioritisation: Designate L10, L18, and L12 as critical-tier lockers requiring priority maintenance contracts and mobile backup capacity protocols. Schedule no planned

closure at these sites without mobile locker deployment to the 13–14 affected exclusive grids.

6.3 Scope for Future Work

Several extensions to the present work are identified as promising directions for future research:

Real demand data: Replacing the synthetic demand dataset with real GPS-traced delivery data from a logistics operator would improve demand accuracy, preference index calibration, and turnover validation. The methodology developed in this project, bi-level MILP, DES simulation, bipartite network analysis, is directly applicable to real data.

Modular Locker Architecture : Move away from fixed-capacity (S/M/L) lockers toward modular units where compartment count scales with local demand, no hard ceiling on parcels stored per site. Study InPost's modular APM design as the reference implementation and assess feasibility of integrating it into the current bi-level MILP framework. Replaces the static capacity constraint with a dynamic sizing variable, fundamentally changes how the upper-level planner model is formulated.

Dynamic demand modelling: The present model uses annual average daily demand. Extending to a stochastic demand model with day-of-week and seasonal variation would improve robustness of the capacity sizing recommendation, particularly for the turnover multiplier.

Machine learning demand forecasting: Using historical transaction data from existing Amazon Locker or Delhivery SmartLocker deployments in Bangalore to train a spatial demand model would reduce reliance on the Gaussian spreading assumption and potentially reveal demand patterns (e.g., catchment decay functions) specific to Indian urban consumers.

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